
ARIMA, VAR & Multivariate Models

ECON 8310: Business Forecasting — Lecture 2

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Lecture Outline

- 1 Stationarity and Differencing
- 2 ARIMA Models
- 3 VAR Models
- 4 Key Takeaways and Roadmap

Stationarity and Differencing

Before fitting any model, the series must have a stable mean and variance.

What Is Stationarity?

Weak (Covariance) Stationarity

A time series $\{y_t\}$ is **weakly stationary** if:

1. $\mathbb{E}[y_t] = \mu$ (constant mean over time)
2. $\text{Var}(y_t) = \sigma^2 < \infty$ (constant, bounded variance)
3. $\text{Cov}(y_t, y_{t-k}) = \gamma_k$ depends only on lag k , not on t

Why it matters:

- If the mean drifts, no single value is forecastable
- If variance grows, prediction intervals become infinite
- Stationarity makes the past informative about the future

Common culprits:

- Trending series (GDP, prices, sales)
- Variance that grows with level

Fix: first-differencing removes a stochastic trend.

Rule of thumb: visible trend or growing variance $\Rightarrow$ likely non-stationary. Test formally with ADF or KPSS.

Unit Roots and Differencing

AR(1): $y_t = \phi_1 y_{t-1} + \varepsilon_t$

Condition	Behavior	Implication
$ \phi_1 < 1$	Shock decays, reverts to mean	Stationary AR(1)
$\phi_1 = 1$	Shock persists <i>permanently</i>	Random walk (unit root)

First differencing removes a stochastic trend: $\Delta y_t = y_t - y_{t-1}$

Testing: ADF (H_0 : unit root) and **KPSS** (H_0 : stationary) — they test opposite nulls; use both.

$d = 0$: already stationary. $d = 1$: one first difference. $D = 1$: seasonal difference ($\Delta_{12}y_t = y_t - y_{t-12}$).

Over-differencing induces negative autocorrelation. Test *after* differencing; stop at $d = 1$ if the result is stationary.

ACF and PACF: Model Identification

Autocorrelation function (ACF):

$$\rho_k = \text{Corr}(y_t, y_{t-k})$$

- MA(q): cuts off after lag q
- AR(p): tails off geometrically

Partial autocorrelation (PACF): correlation between y_t and y_{t-k} controlling for intermediate lags

- AR(p): cuts off after lag p
- MA(q): tails off geometrically

Model	ACF	PACF
AR(p)	Tails off	Cuts off at p
MA(q)	Cuts off at q	Tails off
ARMA	Tails off	Tails off
RW / I(1)	Decays slowly	Big spike at 1

Always unit-root test and difference *before* reading ACF/PACF. The ACF of a non-stationary series is not interpretable.

ARIMA Models

Combine differencing with AR and MA components into one unified model.

ARIMA(p,d,q)

ARIMA(p,d,q) Model

Apply an ARMA(p,q) to the d -times differenced series: $y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \varepsilon_t + \sum_{j=1}^q \theta_j \varepsilon_{t-j}$

Parameters & special cases:

- p : AR lags; d : differences; q : MA lags
- ARIMA(0,1,0): random walk
- ARIMA(0,1,1): equivalent to SES
- ARIMA(1,0,0): AR(1)

Numeric Example

ARIMA(1,1,0) with $\hat{\phi}_1 = 0.5$:
 $y_{100} = 120$, $y_{99} = 118$, $\Delta y_{100} = 2$
 $\hat{y}_{101|100} = 120 + 0.5 \times 2 = \mathbf{121}$

Python: `pmdarima.auto_arima()` searches automatically. `statsmodels.SARIMAX` fits any specified model.

SARIMA: Adding Seasonal Terms

For monthly or quarterly data, autocorrelation often appears at lags $m, 2m, 3m, \dots$

$\text{SARIMA}(p,d,q)(P,D,Q)[m]$ adds seasonal AR and MA polynomials:

Think of SARIMA as two ARIMA models operating at different time scales:

- Non-seasonal part (p, d, q) : captures week-to-week dynamics
- Seasonal part $(P, D, Q)[m]$: captures year-over-year patterns at lag m

Practical workflow:

1. ADF/KPSS $\rightarrow$ choose d
2. Seasonal unit root test $\rightarrow$ choose D
3. ACF/PACF on differenced series $\rightarrow$ guide p, q, P, Q
4. Minimize AIC; check residual ACF (Ljung-Box test)

US Retail Sales

$\text{SARIMA}(1,1,1)(0,1,1)[12]$ is a strong starting point for monthly retail data with trend and December seasonality.

VAR Models

When two series influence each other, model them together symmetrically.

Why Use Multiple Series?

All univariate models (ETS, ARIMA) use only y_t 's own history. But many business variables are *linked*:

- Consumer sentiment today $\rightarrow$ retail spending next month
- Interest rates today $\rightarrow$ housing starts in 3–6 months
- Advertising spend today $\rightarrow$ sales over the next few weeks

If series x_t contains information about y_{t+h} *not already in* y_t 's own history, then including x_t reduces forecast error. **Granger causality** (Granger 1969) formalizes this idea.

VAR: treat all variables symmetrically — each variable predicts all others.

ARIMAX: one direction only — x_t drives y_t but not vice versa.

The VAR(p) Model

For two stationary series y_{1t} and y_{2t} :

$$\begin{cases} y_{1t} = c_1 + a_{11} y_{1,t-1} + a_{12} y_{2,t-1} + \varepsilon_{1t} \\ y_{2t} = c_2 + a_{21} y_{1,t-1} + a_{22} y_{2,t-1} + \varepsilon_{2t} \end{cases}$$

Key ideas:

- a_{12} : how much $y_{2,t-1}$ predicts y_{1t} (what univariate AR misses)
- Each equation estimated by OLS; order p chosen by BIC
- If $a_{12} = a_{21} = 0$: reduces to separate AR models

Parameter explosion: VAR(p) with k variables has $k^2 p$ coefficients.

$k = 5$, $p = 4$: **100 coefficients!**

Keep $k \leq 4$ unless using LASSO-VAR.

Python: `statsmodels.tsa.vector_ar.var_model.VAR`

Use `var.fit(maxlags=p, ic='bic')` for automatic order selection.

Granger Causality Test

Granger Causality

x_t **Granger-causes** y_t if past values of x_t help predict y_t *beyond what y_t 's own past explains.*

Test procedure:

1. Fit VAR with and without x -lags in the y -equation
2. F-test H_0 : all x -lag coefficients = 0
3. Reject $\Rightarrow x_t$ Granger-causes y_t

Business example:

Does consumer sentiment (UMCSENT)

Granger-cause retail sales (RSXFS)?

H_0 : all sentiment lags = 0 in the retail equation.

Impulse response functions (IRFs) trace how a shock propagates through the VAR system over time.

Granger causality $\neq$ structural causality. It tests *predictive content*, not whether x_t mechanically causes y_t .

ARIMAX: Exogenous Variables with ARIMA Errors

When one variable clearly *drives* another (not symmetric), use ARIMAX:

ARIMAX Model

$$y_t = \mathbf{x}_t' \boldsymbol{\beta} + \eta_t, \quad \eta_t \sim \text{ARIMA}(p, d, q)$$

$\mathbf{x}_t$ are exogenous regressors; η_t captures remaining autocorrelation in the residuals.

When to use ARIMAX:

- Causal direction is clear: ads $\rightarrow$ sales
- You already have or can forecast x_t
- Adding x_t reduces forecast error in CV

Python:

```
SARIMAX(y, exog=X,  
order=(p,d,q)).fit()
```

ARIMAX requires a future forecast of x_t to forecast y_t . If that is unavailable, use VAR instead.

Key Takeaways and Roadmap

Stationarity, ARIMA, VAR, and ARIMAX form the classical time series toolkit.

Key Takeaways

1. **Stationarity** is required before fitting time series models. Test with ADF (H_0 : unit root) and KPSS (H_0 : stationary). First-differencing removes a stochastic trend.
2. **ACF** identifies MA order; **PACF** identifies AR order. Always inspect after differencing, not before.
3. **ARIMA**(p, d, q) combines differencing with AR/MA. SARIMA adds seasonal polynomials for periodic autocorrelation. Use `auto_arima` for automatic selection.
4. **VAR** models all variables symmetrically and captures cross-variable dynamics. Keep $k \leq 4$. Use **Granger causality** to test predictive content.
5. **ARIMAX** adds exogenous predictors to ARIMA. Use when causal direction is clear and x_t is forecastable.

Next: Generalized Additive Models (GAMs) — flexible, interpretable nonlinear forecasting with Prophet and pyGAM.

References I



Granger, C. W. J. (1969). "Investigating Causal Relations by Econometric Models and Cross-Spectral Methods". In: *Econometrica* 37.3, pp. 424–438.